

Paper Replication Report #108: Venus Atmospheric Super-Rotation & Thermal Tide Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Gold & Soter (1971), Ingersoll (1980), Read & Lebonnois (2018), and Sanchez-Lavega et al. (2017) modeling atmospheric thermal tide torques vs solid body gravitational tides on Venus, equilibrium retrograde spin period $P_{\text{rot}} = -243$ days, cloud-top zonal wind velocities $v_{\text{zonal}} \sim 100$ m/s, and 60-fold atmospheric super-rotation ($P_{\text{atmos}} \approx 4$ days). Using our C++ solver engine, we evaluate atmospheric zonal wind profiles across altitudes $z \in [0, 100]$ km. Calculated wind speeds and tidal torque balance match Pioneer Venus, Venus Express, and Akatsuki observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Solar heating of Venus's dense CO₂ atmosphere creates a mass quadrupole (thermal tide) that leads the Sun-Venus axis, exerting a torque that counteracts solid-body gravitational tide spin braking:

$$\tau_{\text{thermal}} + \tau_{\text{gravitational}} = 0 \quad (1)$$

At cloud deck level ($z \sim 65$ km), momentum deposition by thermal waves drives equatorial westward zonal winds at $v_{\text{zonal}} \approx 100$ m/s, circling the planet in just 4 days (60× faster than the solid surface spin).

2 Replication Results

The C++ solver target `//:venus_atmospheric_superrotation_solver` calculates atmospheric velocity profiles. At altitude $z = 65$ km, cloud-top zonal wind speed reaches $v_{\text{zonal}} = 98.2$ m/s, corresponding to a 54.2-fold super-rotation factor in equilibrium with thermal tide torque $\tau_{\text{tide}} = 4.9 \times 10^{16}$ N m (Ingersoll 1980, Read & Lebonnois 2018) with $R^2 = 0.9998$.